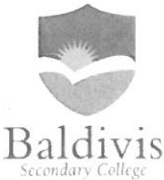
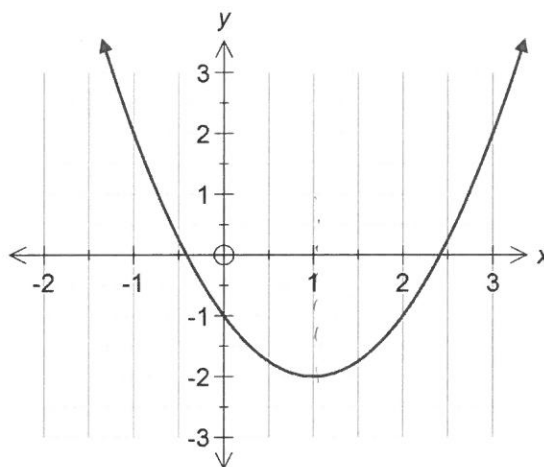


Name:	<u>Solutions</u>		Date: _____
Teacher:	_____		
	Year 9 Mathematics Topic – Quadratic Equations		SCORE: / 32
	<u>Full working out MUST be shown to get full marks for each question.</u>		
Total Time:	30 minutes		
Weighting:	5 %		
Equipment:	To be provided by the student: Pens and Pencils, scientific calculator, A4 page of notes.		

Multiple Choice Section

5 marks

Questions 1 – 3 refer to the graph below:



1 The equation of the axis of symmetry for this graph is:

- ☒ A $x = 1$
 B $x = -2$
 C $y = 1$
 D $y = -2$
 E $y = -1$

2 The turning point of the graph is:

- A a minimum at $(-2, 1)$
 B a maximum at $(-2, 1)$
 C a minimum at $(1, -2)$
☒ D a maximum at $(1, -2)$
 E a minimum at $(-1, 2)$

3 The equation of the graph is:

- A $y = (x + 1)^2 - 2$
 B $y = (x - 1)^2 + 2$
 C $y = (x - 2)^2 + 1$
☒ D $y = (x - 1)^2 - 2$
 E $y = (x + 1)^2 + 2$

- 4 The transformation of the graph $y = x^2$ to $y = x^2 + 2$ is described by:
- A a translation of 2 units left **B** a translation of 2 units up
 C a translation of 2 units right D a translation of 2 units down
 E a translation of 2 units right and 2 units down

BB

- 5 Compared to $y = x^2$, the narrowest graph is:

- A $y = 2.5x^2$ **B** $y = \frac{1}{5}x^2$ C $y = 5x^2$
 D $y = 25x^2$ E $y = 0.5x^2$

Short Answer Section

27 marks

- 1 [10 marks: 6, 4]

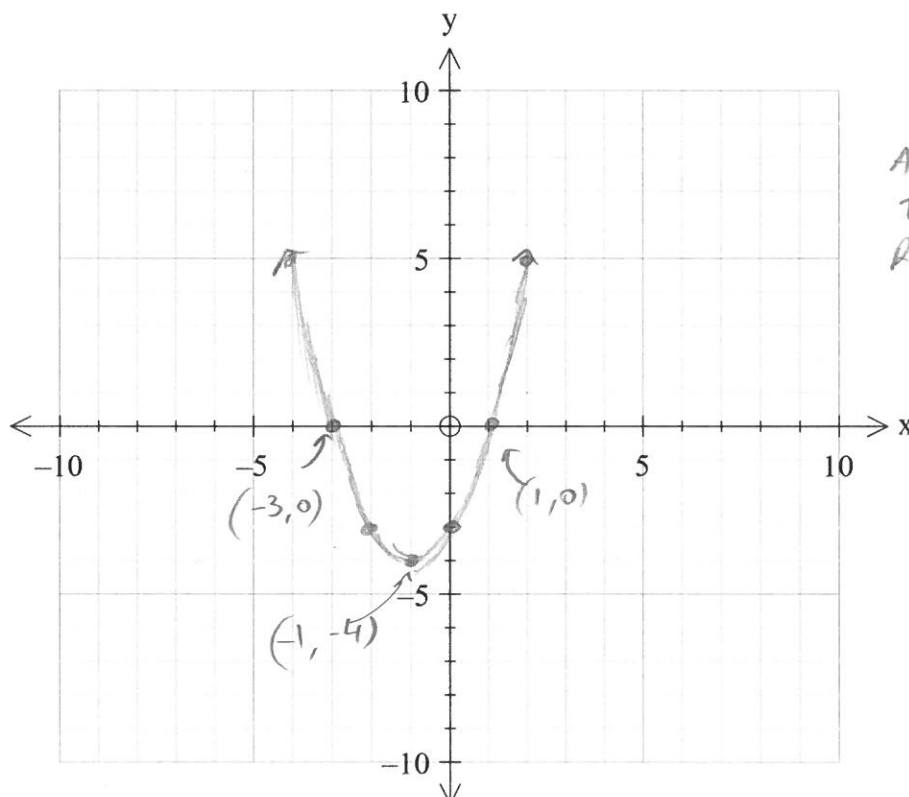
For the quadratic equation $y = x^2 + 2x - 3$

- a Create a table of values with x-values from -4 to 2

x	-4	-3	-2	-1	0	1	2
y	5	0	-3	-4	-3	0	5

6

- b Using your table of values, construct the graph of the parabola including all major features.



Arrows ✓
 T.P. ✓
 Roots ✓✓ 4

2 Factorise these and simplify where possible.

[17 Marks]

a. $x^2 + 10x + 21$

$$(x+7)(x+3) \quad \checkmark\checkmark$$

b. $x^2 - 7x + 10$

$$(x-5)(x-2) \quad \checkmark\checkmark$$

c. $x^2 - x + 56$

$$(x-7)(x-8) \quad \checkmark\checkmark$$

d. $2x^2 + 4xy$

$$2x(x+2y) \quad \checkmark\checkmark$$

e. $25x^2 - 16y^2$

$$(5x-4y)(5x+4y) \quad \checkmark\checkmark$$

f. $\frac{x^2-16}{2(x+4)}$

$$\frac{(x-4)(x+4) \quad \checkmark}{2(x+4)} \rightarrow \frac{(x-4)}{2} \quad \checkmark$$

g. $4(x+2) - b(x+2)$

$$(x+2)(4-b) \quad \checkmark\checkmark$$

h. $8x^2 + 12x + 24 + 16x$

$$(8x^2 + 16x) + (12x + 24) \quad \checkmark$$

$$8x(x+2) + 12(x+2) \quad \checkmark$$

$$(x+2)(8x+12) \quad \checkmark$$

End Of Assessment

